

PERFORMANCE WORK STATEMENT

Mobile Off-road Terrain Measurement System

C.1 GENERAL. The objective of this contract is to provide U.S. Army Aberdeen Test Center (ATC) a stand-alone highly mobile off-road test course and terrain measurement system, consisting of a prime mover with integrated sensor suite, sensor data collection hardware and software. The system shall be called the Profilometer and meet the technical, training, and documentation requirements as described in this document.

C.1.1 SCOPE. This is a non-personal services contract to provide ATC a stand-alone highly mobile off-road test course and terrain measurement system, consisting of a prime mover with integrated sensor suite, sensor data collection hardware and software. The Government shall not exercise any supervision or control over the contract service provider.

C.1.2 BACKGROUND. The test facilities at the U.S. Army Aberdeen Test Center (ATC) contain approximately 64 km (40 mi) of automotive test courses on more than 1320 hectares (3300 acres) of land. Each automotive test course is designed to meet a particular military vehicle specification, and many exceed commercial standards. The courses are divided into two classifications, endurance courses and performance courses. The ATC Profilometer regularly traverses each of the ATC test courses and collects terrain measurements to ensure that they are within specific roughness specifications. Results from data collected from the Profilometer ensure test course consistency over time and provide test course maintenance recommendations.

C.2 APPLICABLE DOCUMENTS AND AVAILABILITY.

C.2.1 APPLICABLE DOCUMENTS.

Refer to TOP 01-1-010 (<https://apps.dtic.mil/sti/pdfs/AD1043586.pdf>) Appendix A for a description of the system to be replaced.

C.2.2 AVAILABILITY OF DOCUMENTS. The government will be responsible for providing the contractor all ATC and/or Department of the Army SOPs or regulations.

C.3 REQUIREMENTS. The ATC Profilometer will be a highly mobile off-road test course and terrain measurement system, consisting of a prime mover with integrated sensor suite, sensor data collection hardware and software. The prime mover will be a rugged off-road capable vehicle. The minimum requirements of the integrated sensor suite are micro-terrain measurement (scanning) instrumentation and an Inertial Navigation System (INS). Additional sensors may be designed into the sensor suite if the contractor determines they add pertinent information to achieve the terrain measurement objective. The INS will be used to determine the position and orientation of the scanning hardware as the prime mover traverses the terrain, allowing the correlation of scan data frames to a common earth-referenced coordinate system. The INS consists of a high-grade Inertial Measurement Unit (IMU) which produces triaxial accelerations and triaxial rotational

measurements, and a differential Global Positioning System (GPS). The Profilometer data collection system will interface with the sensors to produce a terrain scan in post-processing by merging the INS solution with the surface scanning data. The Profilometer will include an end-to-end software package for controlling the data collection system, data processing, and data analysis. Many of the ATC test courses contain very soft soils that become significantly softer when saturated with water. Typical terrain scanning sensors cannot distinguish soil density. The intention of the profilometer is to measure the actual input into the suspension of the vehicles under test. To mitigate the compressibility of the soil, the historical practice only used the single line of data points from behind each wheel for course evaluation. For test courses that have negligible soil compressibility, in which case data outside the wheel paths can be used.

Refer to TOP 01-1-010 Appendix A for a description of the system to be replaced.

The Profilometer shall have a comprehensive software package, which may be composed of one or multiple programs. The program(s) shall perform the sensor initiation and configuration operation, with stages for data collection, merging, reduction and analysis, and results reporting. The programs will be collectively referred to as the software package.

The software package will have user accessible outputs at the different stages of data collection and analysis. At each stage the software must generate outputs in formats which can be read by widely available programs such as MATLAB or Python. The outputs from the different stages will be used for diagnostic and verification purposes, confirming instrumentation functionality and analysis accuracy. The outputs will include:

- Surface scanner outputs
- Calculate the RMS of user selected single line profiles as per TOP 01-1-010
- INS position and orientation solutions
- Differential GPS data
- XYZ points of the merged surface scan
- Distance vs. elevation profiles for selected paths through the surface scan
- Full width Curved Regular Grid (CRG) point cloud output
- Feature identification to include:
 - Potholes (size and count)
 - Washboarding
 - Ruts
 - Amount/location of water on course or in potholes
- A record of any edits made to scan data

The selected contractor will be required to demonstrate how their proposed design meets system requirements in a design review. They will next conduct a demonstration of their prototype in an initial system test and a final system test. The initial system test may occur at the contractor's facility, the government facility, or a third-party facility. The objective of the initial system test demonstration will be to demonstrate the terrain measurement system's ability to collect the appropriate raw data needed to perform the reporting

functions outlined in this document. The final system test shall take place at the government facility where the profiler will scan ATC Endurance and Performance test courses. The profiling system will scan six (6) specified test courses. This data shall be processed using the software package and results will be output in formats meeting the reporting requirements.

C.3.1 PROFILOMETER System Requirements

Each requirement maps to an associated qualification method(s) and has traceability to system requirements. Each requirement includes thresholds and objectives as applicable where:

1. **Threshold (T)** – Minimum Level of performance that is considered achievable at low-to-moderate risk.
2. **Objective (O)** - Level of performance which is the ultimate desired operational goal.

T = O Threshold and Objective are the same requirement level. Therefore, when T = O, there is no contractual obligation to exceed the Threshold requirement.

This section specifies the requirements, that is, those characteristics of the system that are conditions for acceptance. Requirements are sub organized under three overall categories: Prime Mover, Sensors, and Data. Each system element, defined as a group of related requirements, is itemized using a project unique identifier to support testing and traceability. Each system element is annotated by a unique Requirement Identifier (REQ-Category-##). Individual requirements represent constituent capabilities to enable functionality in each system requirement.

Category	Summary	Description
I. Prime Mover Requirements	REQ-PM-01	The prime mover with integrated sensors must be able to traverse difficult off-road terrain T = 25° breakover angle, 27° approach & departure angle, 300mm minimum ground clearance, and 350mm of ground clearance at the breakover point. O = > 26° breakover , > 30° approach & departure angle and >380 mm ground clearance
	REQ-PM-02	The prime mover must be able to traverse secondary and cross-country roads as defined in TOP-01-1-010 and TOP 01-1-011 with integrated sensors T = 4WD with locking transfer case, low range, and front and rear locking differentials T = O
	REQ-PM-03	Prime mover must be protected from course debris and features on secondary and cross-country roads as defined in TOP 01-1-010 and TOP 01-1-011. ()

		T = Prime mover must have comprehensive skid plates protecting the oil pan, transmission, transfer case, fuel tank, and vehicle body T = O
	REQ-PM-04	Prime mover interior must accommodate an operator and trainee in all season conditions T = Cab space for 2 occupants without removing any data collection system equipment. Climate control must be available and operable without compromising power to sensors or operation of the prime mover. Ambient noise in the prime mover must be less than 75 dB while the vehicle is operating. O = Cab space for 3 occupants without removing any data collection system equipment, climate control must be available and operable without compromising power to sensors or operation of prime mover, ambient noise in the prime mover must be < 60 dB.
	REQ-PM-05	Prime mover and mounted equipment must be operable on public roads to move between test courses without special considerations. T = Prime mover complies with all sections of Subtitle 1 of Maryland Title 24 Vehicle Laws with all sensors in place including an overall width of 8 feet or less. T = O
	REQ-PM-06	Prime mover must be cleanable after use in wet, muddy, or dusty conditions T = Exterior of prime mover can be washed in a military wash rack using water at a pressure of 75psi at a flow rate of 25 gallons per minute without ingress. T = O
	REQ-PM-07	The system, including prime mover and sensors, must be operable given storage year round in a non-climate-controlled facility. T = Prime mover must start after up to 3 months storage in a non-climate-controlled facility.

		O = Prime mover must start after more than 3 months storage in a non-climate-controlled facility.
	REQ-PM-08	Prime mover maintenance tasks limited to services provided by ATC fleet maintenance facility. T = Prime mover is a consumer vehicle that has a naturally aspirated gasoline powered engine. T = O
	REQ-PM-09	Prime mover must meet and maintain test course speed requirements for all ATC Endurance and Performance test courses T = Prime mover must traverse the following courses of International Roughness Index (IRI) values (in in/mi) while maintaining these minimum speeds while making valid scans without damage to the equipment. - Paved (70-150 IRI) at 55mph - Secondary Roads (250 -700 IRI) at 35 mph - Cross Country Roads (800 – 1500 IRI) at 20 mph - Trails (5000 + IRI) at 5 mph T = O
	REQ-PM-10	Operators must be able to safely navigate the prime mover in tight difficult terrain. T = Prime mover must include a backup camera if rear visibility from driver seat is impeded. T = O
	REQ-PM-11	Prime mover must have robust recovery provisions. T = Prime mover must include recovery provisions that comply with STANAG 4478 wheeled vehicle towing lugs. T = O
	REQ-PM-12	Prime mover will be fitted with off-road tires capable of traversing ATC terrain. T = Prime mover will be fitted with 35-inch diameter mud terrain tires and will have any necessary driveline or suspension modifications to retain the vehicles performance and warranty including but not limited to axle gearing, springs, shocks, bump stops, fender flares, and wheel offset.

		T = O
II. Sensor Requirements	REQ-SR-01	Controls including arming/ disarming sensors, Graphical User Interface (GUI) for system software control, and Emergency Stop (E Stop) must be operable from driver's seat if required due to sensor safety concerns. T = System arm, disarm, software control GUI, and E Stop must be operable from the driver's seat by a 95th percentile male and a 5th percentile female. T = O
	REQ-SR-02	External warning indicators must be visible to personnel outside of the prime mover when the system is armed. T = Proper signage and indicators IAW ANSI Z136.1 or similar regulatory documents relating to any type of hazardous sensor. T = O
	REQ-SR-03	Scanning instrumentation shall be securely mounted to prime mover. Equipment must stay rigidly in place while traversing off-road courses which induce strong shocks and vibrations. T = The system will be robustly mounted to withstand severe off-road driving as outlined in MIL-STD-810H Large Assembly Transport. The instrumentation and mounting hardware shall not negatively impact off-road performance of prime mover including, but not limited to, approach and departure angles and ground clearance. T = O
	REQ-SR-04	Assembly and disassembly of scanning instrumentation for prime mover maintenance tasks must be completed with reasonable effort. T = Assembly and disassembly of scanning instrumentation that would interfere with prime mover maintenance must be able to be completed by 2 people within a 4-hour window with standard tools, or specialty tools must be provided.

		O = Assembly and disassembly of scanning instrumentation that would interfere with prime mover maintenance must be able to be completed by 1 person within a 4-hour window with standard tools, or specialty tools must be provided.
	REQ-SR-05	Scanning instrumentation must be able to measure continuously over multiple test courses. T = Scanning instrumentation must record for 4 hours continuously. O = Scanning instrumentation must record for > 5 hours continuously.
	REQ-SR-06	Scanning instrumentation must function without a software connection to internet and be transferable to online workstations. T = Scanning instrumentation and software must not require a network connection to operate and will record to removable media. T = O
	REQ-SR-07	The current profilometer uses the points behind the wheel path of the prime mover to calculate course roughness. This compensates for compressible soil that does not significantly affect the suspension. T = Scanning instrumentation must include a mechanism to compensate for compressible soil along the two wheel paths. O = Scanning instrumentation must include a mechanism to compensate for compressible soil across the entire test course width.
	REQ-SR-08	Scanning instrumentation must output a common time basis reference signal at 1 kHz for integration of future equipment and software. T = Scanning system provides an accessible common time basis signal (such as an electrical pulse, TTL, etc.). T = O
	REQ-SR-09	Scanning instrumentation must produce valid scans at slow speeds to allow measurement of the most severe ATC test courses.

		T = Scanning instrumentation produces valid scans at speeds down to 5 mph. O = Scanning instrumentation produces valid scans even if the prime mover stops.
	REQ-SR-10	The processed scan data must compensate for any motion of the mounted sensor relative to the terrain surface. T = Scanning instrumentation must include a differential GPS and inertial navigation system with instrumentation mounted on the scanning sensor platform to correct for platform movement. O = Scanning instrumentation may include additional sensors to compensate for any motion of the mounted sensor relative to the terrain surface.
	REQ-SR-11	Scanning equipment and software must be easily maintainable by operators. T = Operation manuals, parts lists, tool lists, and sources of supply are included at time of delivery. T = O
	REQ-SR-12	The scan width must be sufficient to cover the width of endurance and performance courses at ATC. T = The scan width will be 12 feet or greater. O = The scan width will be variable between 12 and 20 feet.
	REQ-SR-13	Scanning instrumentation must resist ingress of mud, dust, and other environmental hazards common on ATC test courses. T = IP65 compliant. O = IP67 compliant.
	REQ-SR-14	The instrumentation must have sufficient longitudinal resolution at the full range of operational speeds. T = The instrumentation must be able to achieve these longitudinal resolutions at the named speed ranges. - 10 samples per foot between 35-55mph

		- 20 samples per foot between 20-35mph - 30 samples per foot between 10-20 mph - 50 samples per foot below 10mph O = The instrumentation must be able to achieve these longitudinal resolutions at the named speed ranges. - 20 samples per foot between 35-55mph - 30 samples per foot between 20-35mph - 40 samples per foot between 10-20 mph - 60 samples per foot below 10mph
	REQ-SR-15	Scanning instrumentation shall be easily operable by one person in the driver's seat of the prime mover. T = All scanning instrumentation is operable from the driver's seat using a provided GUI. T = O
	REQ-SR-16	Scanning instrumentation must establish a common reference point to monitor test course height changes over months to years. T = Instrumentation must establish a common reference point based on existing features at the government facility. T = O
	REQ-SR-17	The scanning resolution must be sufficient to detect small but relevant course features including ruts, rocks, pebbles, and grout lines between cobblestones. T = The vertical and horizontal precisions must be within 1 mm for relative vehicle position changes less than 3 m at speeds down to 10 mph. O = The vertical and horizontal precisions must be within 0.5 mm for relative vehicle position changes less than 3 m at speeds down to 5 mph.
	REQ-SR-18	The scanning resolution must be sufficient to detect larger course features such as potholes, large rocks, and humps. T = The vertical and horizontal precisions must be within 10 mm for relative vehicle position changes between 3 and 10 m at speeds down to 5 mph.

		O = The vertical and horizontal precisions must be within 5 mm for relative vehicle position changes between 3 and 10 m at speeds down to 5 mph.
	REQ-SR-19	The tolerance of scanning equipment must be sufficient to detect course grades and long-distance features. T = The vertical and horizontal precisions must be within 20 mm for relative vehicle position changes between 10 and 3000 m at speeds down to 5 mph. O = The vertical and horizontal precisions must be within 10 mm for relative vehicle position changes between 10 and 3000 m at speeds down to 5 mph.
	REQ-SR-20	Safety equipment required for safe operation of the system must be included. T = Safety equipment including but not limited to any personal protective equipment and covers for any hazardous sensors. O = T
	REQ-SR-21	The scanning hardware must have sufficient measurement range to capture all the large obstacles on the test courses. T = The vertical scanning range must be at least 508 mm for speeds below 25 mph and at least 254 mm for speeds above 25 mph. O = The vertical scanning range must be greater than 508 mm for speeds below 25 mph and greater than 254 mm for speeds above 25 mph.
III. Data Requirements	REQ-DR-01	Data recorded or reduced by provided software is exportable in a non-proprietary format to be readable by other software including Python or MATLAB. T = All data recorded is exportable in a non-proprietary format such as .csv. T = O
	REQ-DR-02	Data will be visualized and processed within a quick turnaround time.

		T = Data visualization, processing, and result reporting must not take longer than 24 hours to complete from the end of scanning to the production of the final reports. O = Data visualization, processing, and result reporting must not take longer than 10 hours to complete from the end of scanning to the production of the final reports.
	REQ-DR-03	Data from inertial measurement instrumentation must be used to correct scan data for vehicle motions. T = Software is provided to post-process GNSS corrections, combine GNSS and inertial measurements into an integrated position and orientation solution, and translate scan sensor data to earth-referenced coordinates. T = O
	REQ-DR-04	Data recorded by the Profilometer produces equivalent metrics to the current ATC Laser Profiler. T = Data collected includes: - GPS Time - Latitude - Longitude - Ellipsoidal height - Northing - Easting - Roll Angle - Pitch Angle - Kappa Angle - Heading Angle - Euler Angles - Distance from scanning equipment to ground - X, Y, and Z displacement coordinates of prime mover compared to a reference point - X, Y, and Z coordinates measured by scanning instrumentation O = Additional data is provided.
	REQ-DR-05	Users must be able to visualize data to make corrections for errors caused by difficult terrain conditions such as deep water, heavy mud, and dust.

		T = Using provided software or tools, users can visualize and correct erroneous measurements from the full scan. T = O
	REQ-DR-06	Provided software must be able to generate 3D renderings of course scans capable of being viewed and edited. T = 3D renderings can be made of full test course width and at least 100 ft in length. O = 3D renderings can be made of full test course width and full test course length.
	REQ-DR-07	Provided software must allow for automated or user defined detection of terrain features. T = Software can identify and count potholes, wash boarding, ruts, and bumps in test courses using user defined parameters such as height, depth, length, width, and shape. O = Software can identify and count potholes, wash boarding, ruts, and bumps in test courses using user defined parameters such as height, depth, length, width, and shape. Software can detect if potholes or other test course features are filled with water.
	REQ-DR-08	Provided software for data analysis and reporting must be Army Golden Master (AGM) Compliant. T = All provided software is AGM Compliant. T = O
	REQ-DR-09	Provided software shall have features to indicate that data is being collected and have verification tools to preliminarily check data in the field. T = Data collection state, including errors that impede accurate data collection, INS solution health, and scanner activity shall be indicated on the GUI at the driver's seat. T = O

C.3.2 PROFILOMETER System Validation

C.3.2.1 Systems Requirements review and Final Course Walkthrough

Subsequent to the contract award, the selected vendor shall attend a post-award conference at the designated government facility. This conference will include a site survey of the test courses and a formal System Requirements Review. The purpose of this review is to validate and finalize all system requirements, ensuring a comprehensive and mutual understanding is established between the Government and the vendor prior to the commencement of work. Furthermore, the contractor shall be required to draft and present a detailed milestone plan for government review and approval during this conference. This milestone plan shall include, at a minimum, projected dates for all reviews, tests, and training events as identified in the Performance Work Statement (PWS), section 3.2.

C.3.2.2 Design Review

This is a technical review to ensure that a system can proceed into fabrication, demonstration, and test that can meet stated performance requirements. The vendor shall conduct the design review. The design review assesses the system final design as captured in product specifications and will ensure that the system has been captured in detailed design documentation. The design review shall be scheduled by the vendor prior to the beginning of the integration of the scanning system into the prime mover. Areas of focus will be:

C.3.2.2.1 Review of the detailed specifications of selected sensors' performance parameters.

C.3.2.2.2 Review of the detailed interface specifications of selected sensors and the recording system.

C.3.2.2.3 Review the detailed sensor integration plan (if any) for the selected sensors' data.

C.3.2.2.4 Review the detailed sensor to prime mover integration plan.

C.3.2.2.5 Review a detailed user interface description.

C.3.2.2.6 Review the selection of the prime mover.

C.3.2.3 Initial System Test

The initial system test may be conducted at a government facility, vendor facility, or third-party facility. The system shall measure a variety of obstacles to verify that it can accurately capture the 3D profile of test course features. The software shall be able to control the system in the prime mover and process collected data. Scan results shall be compared with existing measurements. Areas of focus will be:

C.3.2.3.1 Demonstration of the sensors' performance parameters.

C.3.2.3.2 Demonstration of data communications from sensors to the recording system.

C.3.2.3.3 Demonstration of any sensor integration data results.

C.3.2.3.4 Review of integrated sensors on prime mover system.

C.3.2.3.5 Demonstration of the user interface operation.

C.3.2.3.6 Demonstration of the data processing tools.

C.3.2.3.7 Evaluation of the output information after data processing.

C.3.2.3.8 Demonstration of the prime mover operating over the terrain.

C.3.2.4 Final System Test

The final system test will be conducted on ATC test courses. The system will measure 2 test courses each of primary, secondary, and cross-country terrain with a variety of permanent obstacles including but not limited to half rounds, potholes, and ATC Profile IV. The results will be compared to existing measurements and course profiles to ensure the scanning is accurate and all required data is capturable. To ensure the system hardware is robust enough to survive many years and miles of off-road courses the vehicle will perform 5 laps of ATC vibration courses as per MIL-STD-810H Large Assembly Transport without damage to the equipment. Areas of focus will be:

C.3.2.4.1 Verification of sensor performance parameters.

C.3.2.4.2 Verification of correct data communications from sensors to the recording system.

C.3.2.4.3 Verification of sensor integration data results.

C.3.2.4.4 Review of integrated sensors on prime mover system.

C.3.2.4.5 Verification of user interface operation.

C.3.2.4.6 Evaluation data processing tools.

C.3.2.4.7 Evaluation of resultant information after data processing process.

C.3.2.5 Training

The vendor shall develop and provide system operation and maintenance training through a combination of classroom, written instructions, and/or hands-on operation. The vendor shall develop and conduct training for users/operators and local maintainers to understand the functional and operational capabilities. The vendor shall provide local support maintenance training and training system documents required to support troubleshooting and maintenance, diagnostics, fault isolation, calibration, adjustments, removal and replacement procedures, use of built in test, and repair, and other tasks associated with local support maintenance.

C.3.2.6 Warranty

For a 12-month period after final delivery and acceptance of the system. The vendor will provide warranty support on all equipment and software necessary to maintain the system.

C.4 DELIVERABLES AND DELIVERY SCHEDULE:

Vendor shall deliver a highly mobile off-road test course and terrain measurement system, consisting of a prime mover with integrated sensor suite, data collection hardware and software. Vendor shall deliver complete documentation of the system, system operations, software instructions and system maintenance. Vendor shall supply

the system 12 months after contract award.

Deliver to Dominic Fonseca
US Army Aberdeen Test Center
6850 Lanyard Road, Bldg 358
Aberdeen Proving Ground, Maryland 21005-5059
520-708-0940
dominic.r.fonseca.civ@army.mil

C.5 GOVERNMENT FURNISHED FACILITIES, EQUIPMENT, AUTOMATION SYSTEMS, MATERIALS AND SERVICES

None

C.6 CONTRACTOR-FURNISHED FACILITIES, EQUIPMENT, AUTOMATION SYSTEMS, MATERIAL AND SERVICES

The contractor shall provide all facilities, equipment, automation systems, materials and services required for satisfaction of the contract terms.

C.7 QUALITY ASSURANCE SURVEILLANCE PLAN (QASP)

Attached as a separate document.

C.8 CONTRACTING OFFICER'S REPRESENTATIVE (COR) AND TECHNICAL POINT OF CONTACT (TPOC):

C.9 NON-PERSONAL SERVICE STATEMENT

Contractor employees performing services under this order will be controlled, directed, and supervised at all times by management personnel of the contractor. Contractor management will ensure that employees properly comply with the performance work standards outlined in the statement of work. Contractor employees will perform their duties independent of, and without the supervision of, any Government official. The tasks, duties, and responsibilities set forth in the task order may not be interpreted or implemented in any manner that results in any contractor employee creating or modifying Federal policy, obligating the appropriated funds of the United States Government, overseeing the work of Federal employees, providing direct personal services to any Federal employee, or otherwise violating the prohibitions set forth in Parts 7.5 and 37.1 of the Federal Acquisition Regulation (FAR).

C.10 Data Rights Assertion and Software License Agreement

The United States Government shall own all rights, title and interest in any and all data created, collected, originated, stored, generated, or transmitted, as well as all data that is the output of any processing or manipulation of said data, in the course of the use of this device, performing its function. All rights, title, and interest to any and all data acquired through the use of this device shall be the sole property of the United States Government. Additionally, the Contractor shall not access, collect, store, copy, disclose, or otherwise use the Data for any purpose, other than what is strictly necessary to fulfill its obligations under the contract. The contractor is expressly prohibited from analyzing or otherwise using the data in any fashion, for its own purposes, including but not limited to commercial purposes, research, or internal business purposes, or for the benefit of any third party. Upon termination or expiration of this contract, the Contractor shall destroy any data in its possession or control and provide a written certification of same. This clause shall take precedence over any license agreement subject to the use of software related to the use of this device, which contradicts or contravenes the above terms and conditions. Nothing in this clause shall be deemed to be construed as an assertion by the United States Government to any rights, title, or interest, in any proprietary data, or materials, documentation, work product, software and software modifications, or other intellectual property developed by the contractor in the manufacture of the device.